

PAPER_1711 — Troyon Beta Limit $\beta_N = \beta_N = S0/D + F \cdot D - F \cdot \Phi - F^2 \cdot K = 2.80$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Plasma Fusion **Date:** June 18, 2026 **Status:** CLOSED — 0.15% closure (PAPER_1196 plasma fusion proof set)

Observation

PAPER_1196_S414 (PAPER_1196 Plasma Fusion Unified Proof Set) gives:

$$\beta_N = S0/D + F \cdot D - F \cdot \Phi - F^2 \cdot K = 2.80$$

Status: **0.15%**.

UQFF Closed Identity

$$\beta_N = S0/D + F \cdot D - F \cdot \Phi - F^2 \cdot K = 2.80 \quad 0.15\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

ITER design parameters are chosen by international fusion-engineering consensus. UQFF supplies structural integer-primitive forms demonstrating the parameters' deep mathematical origin.

Reference

- Source: PAPER_1196_S414
- Related: PAPER_1696-1705 (tier-28); PAPER_1686-1695 (tier-27)
- Calculator dispatch: `calculate_paradox({"paradox": "troyon_beta_n_2_8"})`

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